

Finite Saturation Collapse Characterizes Frechet Differentiability of Convex Molecule Series

A full positive answer to Question 1 of arXiv:2003.01439

Automated mathematics run `fa_banach_001`, lane 2

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Status. Candidate full solution, likely valid, pending human review. For a given norm-one convex series of molecules, the theorem below gives an exact metric criterion for Frechet differentiability. The source’s separate Question 2 was answered negatively in later literature and is recorded in Section 7.

1 Source questions

Let $(M, d, 0)$ be a pointed metric space. The Lipschitz-free space $\mathcal{F}(M)$ is the canonical predual of $\text{Lip}_0(M)$, where functions in $\text{Lip}_0(M)$ vanish at 0 and carry the Lipschitz norm. For distinct $x, y \in M$, write

$$m_{xy} = \frac{\delta_x - \delta_y}{d(x, y)}.$$

Aliaga and Rueda Zoca [1] studied norm-one series

$$\mu = \sum_{i=1}^{\infty} \lambda_i m_{x_i y_i}, \quad \lambda_i > 0, \quad \sum_{i=1}^{\infty} \lambda_i = 1. \quad (1)$$

They obtained a metric characterization of Gateaux differentiability and a Frechet characterization in the finitely supported case. They ended with the following questions (p. 15 of the arXiv PDF):

Question 1 asks for a metric characterization, in the infinite-support case, of those limits of convex series that are points of Frechet differentiability. We give such a characterization using only finite distance data and the McShane–Whitney extension formulas.

It is known, and recalled in the source, that $\mathcal{F}(M)$ can possess a point of Frechet differentiability only if M is bounded and uniformly discrete. We therefore work under precisely these necessary ambient hypotheses. Put

$$\theta = \inf_{s \neq t} d(s, t) > 0, \quad D = \text{diam}(M) < \infty. \quad (2)$$

This shows that $x \otimes x$ is not a point of Gâteaux differentiability in spite of the fact that x is a point of Gâteaux differentiability. The authors are grateful to Ginés López-Pérez for pointing out this example to them.

Let us finish this section with two open questions. First, we have obtained in Theorem 3.1 a metric characterisation of those limits of convex series of molecules which are points of Gâteaux differentiability. However, in the case of Fréchet differentiability we have only characterised elements of finite support.

Question 1. Is there any metric characterisation of limits of convex series with infinite support that are points of Fréchet differentiability?

Second, all our work on points of Fréchet differentiability has focused on limits of convex series of molecules, as this allows us to consider a fixed set of points in the metric space M . However, the following question makes sense.

Question 2. Let M be a uniformly discrete and bounded metric space. Is there any point of Fréchet differentiability $\mu \in \mathcal{F}(M)$ that is not the limit of a convex series of molecules?

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REFERENCES

Figure 1: The complete context and text of Questions 1 and 2 in the source.

2 Proof intuition

Saturating the first n molecules cuts out a finite-codimensional face E_n of the dual Lipschitz unit ball. The extremal McShane–Whitney extensions compute the uniform diameter of this face exactly from metric data; the resulting number is Δ_n . If μ is Frechet smooth, the coefficient tail places E_n in an increasingly thin norming slice, so Smulyan’s lemma forces Δ_n to vanish.

For the converse, a function in a thin norming slice almost saturates each of the first finitely many molecules. Compactness of the corresponding finite Lipschitz polytope lets us correct those values to exact saturation. Clamping the original function between the corrected lower and upper metric extensions makes this a global correction. Uniform discreteness upgrades its small uniform error to small Lipschitz-norm error. Thus the entire thin slice lies near the collapsing face E_n , which is precisely Smulyan’s criterion for Frechet differentiability.

3 The finite metric invariant

Fix a series (1) with $\|\mu\| = 1$. For $n \geq 1$ let

$$S_n = \{0, x_1, y_1, \dots, x_n, y_n\},$$

viewed as a set, and define the finite saturated-potential polytope

$$P_n = \left\{ a : S_n \rightarrow \mathbb{R} : \begin{aligned} &a(0) = 0, \quad |a(s) - a(t)| \leq d(s, t) \quad (s, t \in S_n), \\ &a(x_i) - a(y_i) = d(x_i, y_i) \quad (1 \leq i \leq n) \end{aligned} \right\}. \quad (3)$$

This set is nonempty. Indeed, take a norming $f \in S_{\text{Lip}_0(M)}$ for μ . Since $f(m_{x_i y_i}) \leq 1$ and all coefficients in (1) are positive, equality

$$1 = f(\mu) = \sum_i \lambda_i f(m_{x_i y_i})$$

forces $f(m_{x_i y_i}) = 1$ for every i . Thus $f|_{S_n} \in P_n$.

For $a \in P_n$, define its lower and upper metric envelopes on M by

$$L_a(z) = \max_{s \in S_n} (a(s) - d(s, z)), \quad U_a(z) = \min_{s \in S_n} (a(s) + d(s, z)). \quad (4)$$

Both are normalized 1-Lipschitz extensions of a . Define

$$\Delta_n = \sup_{a, b \in P_n} \sup_{z \in M} (U_a(z) - L_b(z)). \quad (5)$$

The number Δ_n is nonnegative and finite. It is determined entirely by the finite distance matrix on S_n , the distances from an arbitrary point z to S_n , and the first n oriented pairs in the representation.

4 Why the invariant is the right one

Consider the decreasing sequence of saturated faces

$$E_n = \{g \in B_{\text{Lip}_0(M)} : g(x_i) - g(y_i) = d(x_i, y_i) \text{ for } 1 \leq i \leq n\}. \quad (6)$$

Every member of E_n has norm one. The next lemma translates the metric quantity (5) exactly into the uniform diameter of this face.

Lemma 1 (Exact envelope diameter). *For every $n \geq 1$,*

$$\text{diam}_{\|\cdot\|_\infty}(E_n) = \Delta_n. \quad (7)$$

Consequently,

$$\Delta_n \leq D \text{diam}_{\|\cdot\|_L}(E_n), \quad \text{diam}_{\|\cdot\|_L}(E_n) \leq \frac{2}{\theta} \Delta_n. \quad (8)$$

Proof. If $g \in E_n$ and $a = g|_{S_n}$, then $a \in P_n$ and the extension inequalities give

$$L_a(z) \leq g(z) \leq U_a(z) \quad (z \in M).$$

Thus, for $g, h \in E_n$, with restrictions $a, b \in P_n$,

$$g(z) - h(z) \leq U_a(z) - L_b(z) \leq \Delta_n.$$

Swapping g and h deals with the absolute value, so the left side of (7) is at most Δ_n .

Conversely, $U_a \in E_n$ and $L_b \in E_n$ for every $a, b \in P_n$. Therefore every value $U_a(z) - L_b(z)$ appearing in (5) is a pointwise difference of two members of E_n . Taking suprema proves the reverse inequality.

Finally, for every $u \in \text{Lip}_0(M)$, normalization at 0 and (2) give

$$\|u\|_\infty \leq D\|u\|_L, \quad \|u\|_L \leq \frac{2}{\theta}\|u\|_\infty. \quad (9)$$

Apply these inequalities to differences of members of E_n and use (7). $\square$

The reverse implication in the main theorem requires the fact that almost saturated data on a fixed finite set can be corrected to exactly saturated data. The useful point is that no quantitative polyhedral error bound is needed.

Lemma 2 (Finite-face correction). *Fix n . Let*

$$Q_n = \{v : S_n \rightarrow \mathbb{R} : v(0) = 0, |v(s) - v(t)| \leq d(s, t) \text{ for } s, t \in S_n\}.$$

For $\eta \geq 0$ set

$$\omega_n(\eta) = \sup\left\{\text{dist}_\infty(v, P_n) : v \in Q_n, \begin{array}{l} 0 \leq d(x_i, y_i) - [v(x_i) - v(y_i)] \leq \eta \\ \text{for } 1 \leq i \leq n \end{array}\right\}. \quad (10)$$

Then $\omega_n(\eta) \rightarrow 0$ as $\eta \downarrow 0$.

Moreover, if $g \in B_{\text{Lip}_0}(M)$, $v = g|_{S_n}$, and $\text{dist}_\infty(v, P_n) \leq \rho$, then there is $h \in E_n$ such that

$$\|g - h\|_\infty \leq \rho, \quad \|g - h\|_L \leq \frac{2\rho}{\theta}. \quad (11)$$

Proof. The polytope Q_n is compact because it is a closed bounded subset of the finite-dimensional space of normalized real data on S_n . If the first assertion failed, there would be $\varepsilon > 0$, numbers $\eta_k \downarrow 0$, and $v_k \in Q_n$ whose displayed deficits are at most η_k , but with $\text{dist}_\infty(v_k, P_n) \geq \varepsilon$. Pass to a convergent subsequence, $v_k \rightarrow v \in Q_n$. Every saturation deficit of v is zero, so $v \in P_n$, contradicting the distance bound.

For the second assertion, choose $a \in P_n$ with $\|a - v\|_\infty \leq \rho$; compactness allows the minimizing choice. Denote by L_v, U_v the envelope formulas (4) formed from v . Since g extends v ,

$$L_v \leq g \leq U_v.$$

The envelope formulas also show pointwise that

$$|L_a - L_v| \leq \rho, \quad |U_a - U_v| \leq \rho.$$

Now clamp g between the new envelopes:

$$h(z) = \min\{U_a(z), \max\{L_a(z), g(z)\}\}. \quad (12)$$

The minimum or maximum of real-valued 1-Lipschitz functions is again 1-Lipschitz. Also $L_a = U_a = a$ on S_n , hence $h|_{S_n} = a$ and $h \in E_n$. The two envelope estimates show that the clamping moves g pointwise by at most ρ . The second bound in (11) follows from (9) applied to $g - h$. $\square$

5 Main characterization

Theorem 1 (Finite saturation collapse criterion). *Let $(M, d, 0)$ be bounded and uniformly discrete, and let μ have a norm-one convex-series representation (1). Form P_n, L_n, U_n , and Δ_n from (3)–(5). Then*

$$\boxed{\mu \text{ is a point of Frechet differentiability of } \mathcal{F}(M) \iff \lim_{n \rightarrow \infty} \Delta_n = 0.} \quad (13)$$

Equivalently, the common saturated extensions of longer finite prefixes have uniformly collapsing diameter.

Proof. We use the following form of Smulyan's lemma: $\mu \in S_{\mathcal{F}(M)}$ is a point of Frechet differentiability if and only if

$$\text{diam}_{\|\cdot\|_L}\{g \in S_{\text{Lip}_0}(M) : g(\mu) > 1 - \alpha\} \longrightarrow 0 \quad (\alpha \downarrow 0). \quad (14)$$

Suppose first that μ is a point of Frechet differentiability. Put $r_n = \sum_{i>n} \lambda_i$. If $g \in E_n$, then $g(m_{x_i y_i}) = 1$ for $i \leq n$ and $g(m_{x_i y_i}) \geq -1$ for $i > n$. Hence

$$g(\mu) \geq \sum_{i \leq n} \lambda_i - \sum_{i > n} \lambda_i = 1 - 2r_n > 1 - 3r_n. \quad (15)$$

Because all coefficients are positive and the series is infinite, $r_n > 0$, while $r_n \rightarrow 0$. Thus E_n lies in the $3r_n$ -slice from (14). Its Lipschitz-norm diameter tends to zero, and (8) gives $\Delta_n \rightarrow 0$.

Conversely, suppose $\Delta_n \rightarrow 0$. Fix $\varepsilon > 0$. By the exact diameter lemma and (8), choose n so large that

$$\text{diam}_{\|\cdot\|_L}(E_n) < \frac{\varepsilon}{3}. \quad (16)$$

Keep this n fixed and set

$$\lambda_* = \min_{1 \leq i \leq n} \lambda_i > 0.$$

Let $g \in S_{\text{Lip}_0}(M)$ satisfy $g(\mu) > 1 - \delta$. Since $g(m_{x_i y_i}) \leq 1$ for every i ,

$$0 \leq \sum_i \lambda_i [1 - g(m_{x_i y_i})] = 1 - g(\mu) < \delta. \quad (17)$$

In particular, for $1 \leq i \leq n$,

$$\begin{aligned} 0 &\leq d(x_i, y_i) - [g(x_i) - g(y_i)] \\ &= d(x_i, y_i)[1 - g(m_{x_i y_i})] < \frac{D\delta}{\lambda_i} \leq \frac{D\delta}{\lambda_*}. \end{aligned} \quad (18)$$

The finite-face correction lemma therefore provides $\hat{g} \in E_n$ with

$$\|g - \hat{g}\|_L \leq \frac{2}{\theta} \omega_n \left(\frac{D\delta}{\lambda_*} \right). \quad (19)$$

As $\delta \downarrow 0$, the right side tends to zero. Choose $\delta > 0$ so small that it is less than $\varepsilon/3$.

If g_1, g_2 are any two members of this δ -slice, correct them to $\hat{g}_1, \hat{g}_2 \in E_n$. By (16) and (19),

$$\|g_1 - g_2\|_L \leq \|g_1 - \hat{g}_1\|_L + \|\hat{g}_1 - \hat{g}_2\|_L + \|\hat{g}_2 - g_2\|_L < \varepsilon.$$

The dual-sphere slice diameter therefore tends to zero. By (14), μ is a point of Frechet differentiability. $\square$

Remark 1 (Representation and order). *The test is stated for the given convex-series representation, as requested in the source question. If the same vector has another positive norm-one convex-series representation, the criterion has the same truth value for that representation, because Theorem 1 identifies it with the representation-independent property of Frechet differentiability. The same argument also handles repeated points and repeated pairs: S_n is a set, and compatibility of all constraints follows from the common norming function used after (3).*

6 Sanity checks

The infinite star. Let $M = \{0, 1, 2, \dots\}$ with $d(0, k) = 1$ and $d(j, k) = 2$ for distinct positive integers, and let $\mu = \sum_{k \geq 1} 2^{-k} \delta_k$. This is the source's ℓ_1 example. After saturation of the first n molecules, the values at every $k > n$ may still range independently from -1 to 1 . Hence $\Delta_n = 2$ for every n , correctly detecting the failure of Frechet differentiability.

Finite support. For a finite convex combination, the decreasing family E_n stabilizes at the common norming face. Collapse says exactly that this face is a singleton. Together with equivalence of the uniform and Lipschitz norms, this recovers the source's finite-support Frechet criterion.

The source’s positive infinite example. In Example 3.7 of [1], the points are arranged so that saturation of longer prefixes traps all remaining point values in intervals of width tending to zero. In the present language this is precisely $\Delta_n \rightarrow 0$, matching the proved Frechet differentiability of that example.

7 The second question and later literature

Question 2 asks whether a bounded uniformly discrete M can admit a point of Frechet differentiability in $\mathcal{F}(M)$ which is not a limit of a convex series of molecules. Aliaga, Pernecka, and Smith later proved the stronger statement that every element of $\mathcal{F}(M)$ is a convex series of molecules whenever M is uniformly discrete [2]. Thus the answer to Question 2 is *no*; boundedness and Frechet differentiability are not needed for that conclusion. This is a literature answer, not part of the new proof of Theorem 1.

8 Verification and novelty boundary

The proof has two independently checkable components. First, the McShane–Whitney formulas make Δ_n exactly the uniform diameter of the finite saturated face, not merely an upper bound. Second, compactness of the finite data polytope converts approximate saturation in a thin dual slice into proximity to that exact face. Uniform discreteness then upgrades the uniform correction to Lipschitz-norm correction. These yield both directions of Smulyan’s slice criterion.

Cheap run indexes were searched for the arXiv id, title, and core terms. Bounded exact-phrase and close-variant searches found the later answer to Question 2 but no answer to Question 1, including no finite-saturation or extension-polytope formulation. This is not an exhaustive priority search; novelty confidence is moderate. Human review should focus on the finite-face correction, the exact envelope-diameter identity, and possible equivalent terminology in the literature.

References

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